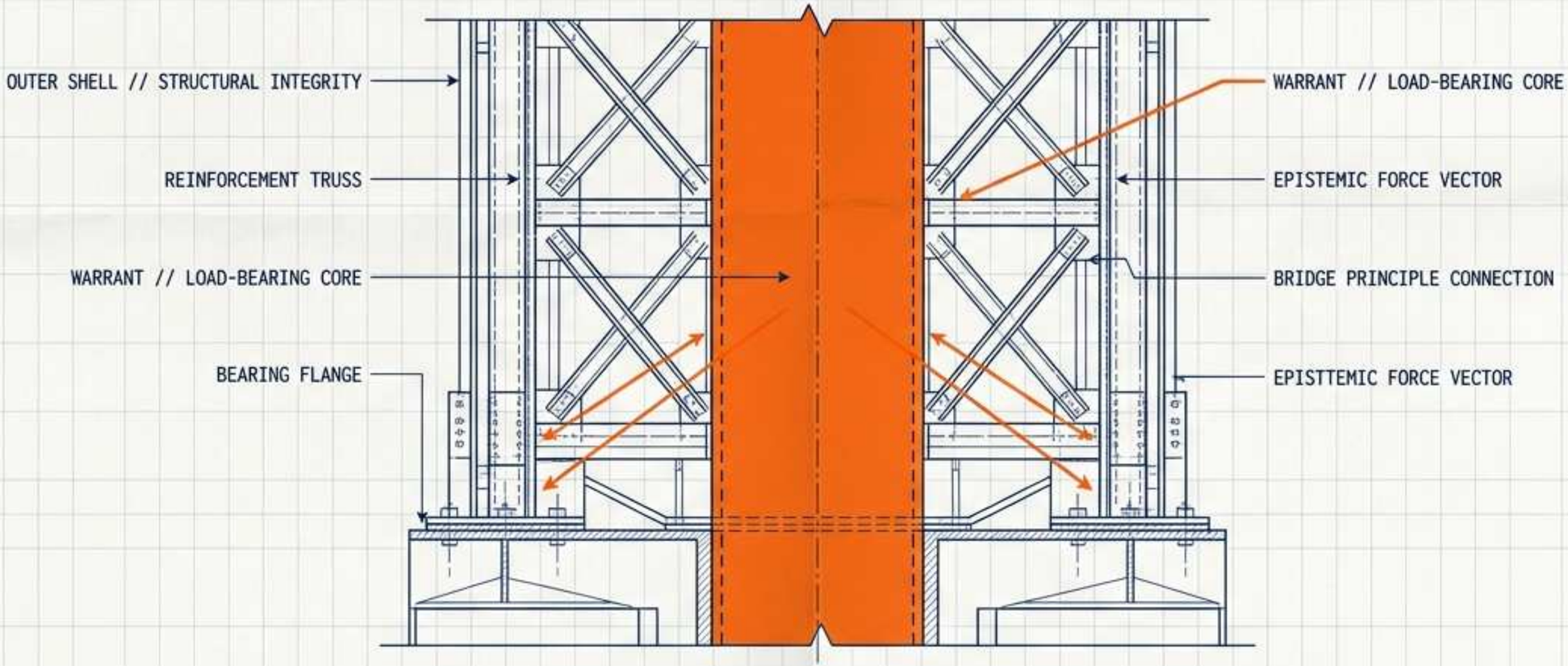
STRUCTURAL HONESTY

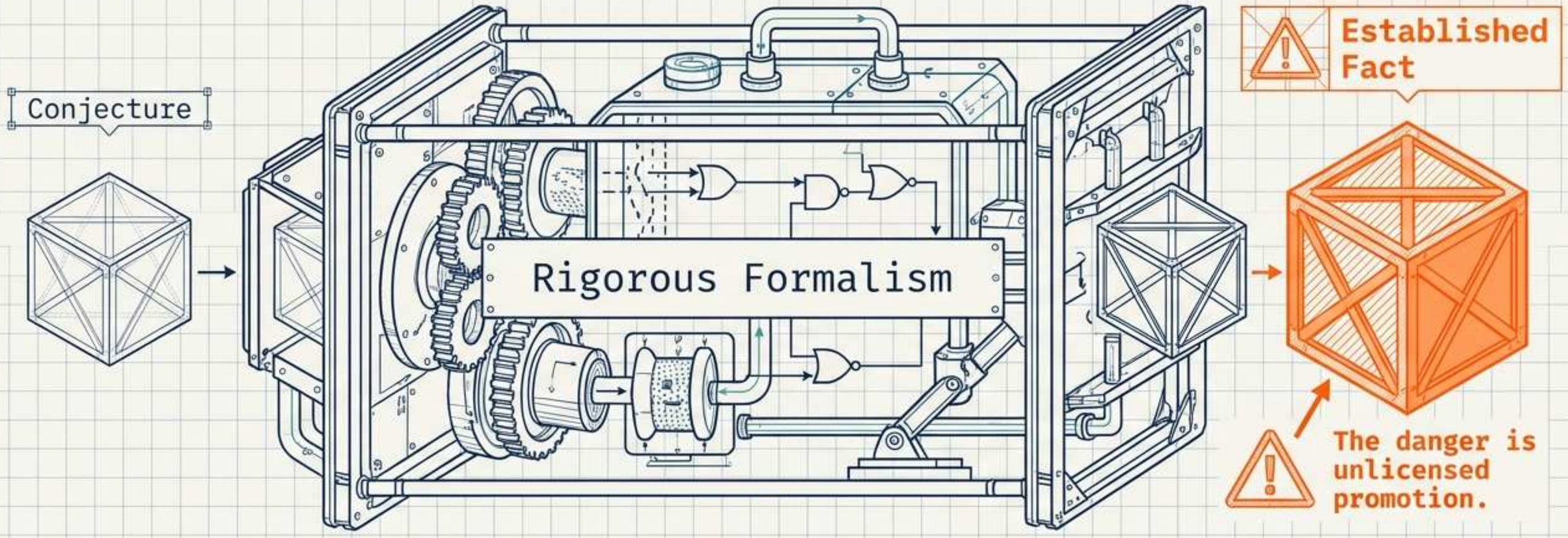
The Preservation of Recoverable Warrant Across Transformations.



INITIATING THE EPISTEMIC BLUEPRINT // SCALE: MACRO TO MICRO

NotebookLM

The Danger of Epistemic Laundering



A correct local operation, embedded in an otherwise rigorous structure, can quietly inherit a strength it was never separately shown to possess.

A numerical observation does not become an analytic result simply because the agreement is striking.

The danger is not ordinary falsehood. The danger is structural.

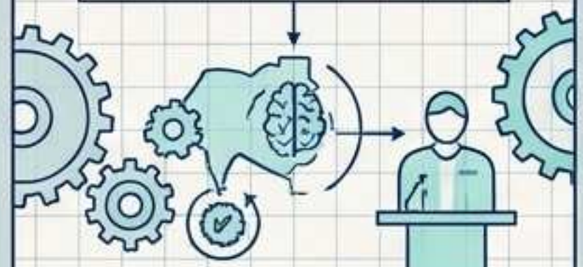
NotebookLM

The Integrity Triad: A Structural Distinction

Sincerity

Domain: Agent vs. Belief

Does the speaker believe it?

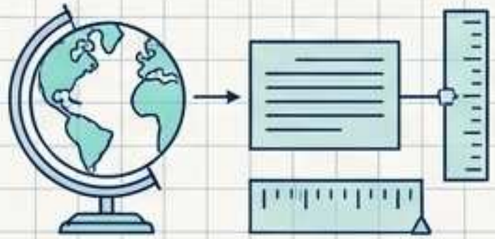


Failure Mode: Deception

Factual Accuracy

Domain: Statement vs. World

Is this empirically correct?

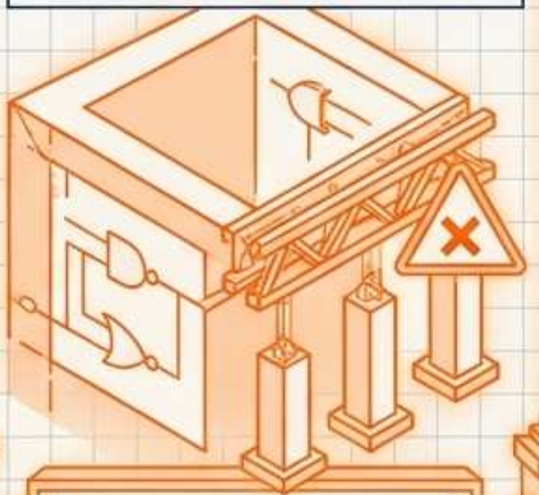


Failure Mode: Error

Structural Honesty

Domain: Representation vs. Warrant

Does this claim exceed the warrant of its underlying structure?



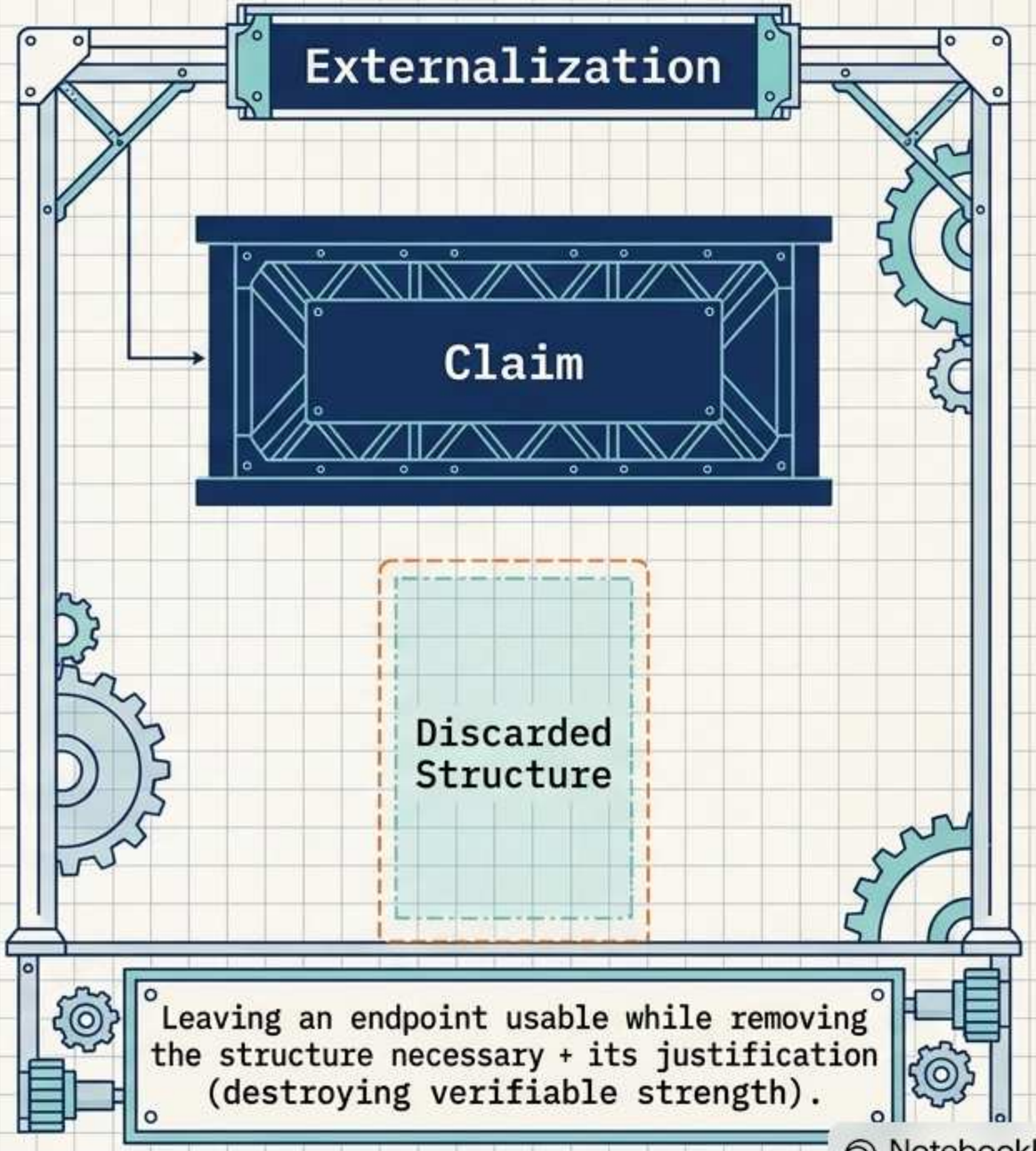
Failure Mode: Unlicensed Promotion

Takeaway: Structural dishonesty does not require deception, and it does not require that any individual statement be false.

NotebookLM

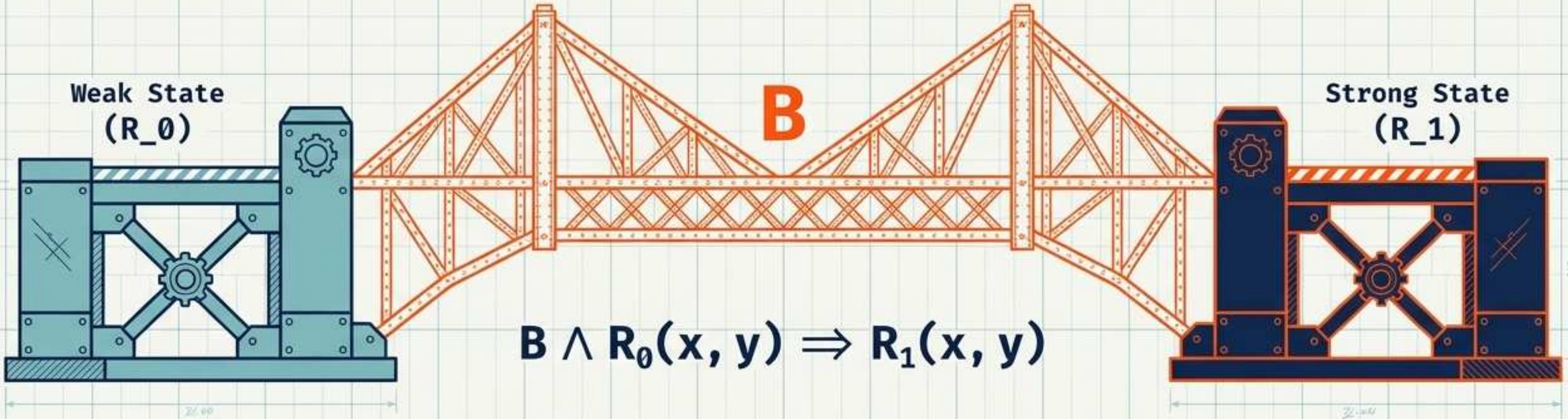
The Two Modalities of Structural Failure





NotebookLM

Bridge Principles & The Warrant Budget



$$B \wedge R_0(x, y) \Rightarrow R_1(x, y)$$

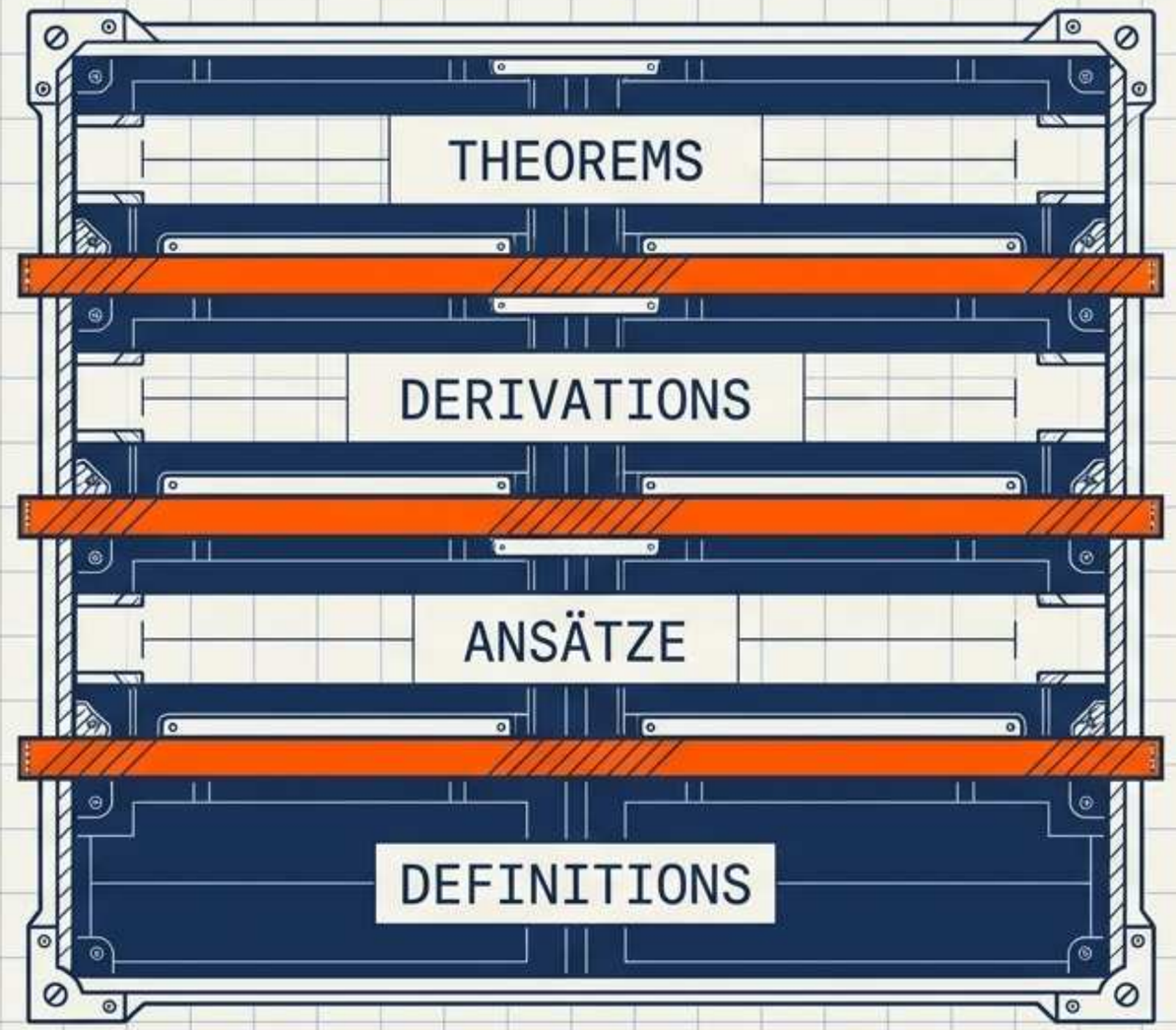
The Warrant Budget Heuristic

$$W(T_x) \leq W(x)$$

- A purely representational transformation (T) cannot increase epistemic entitlement simply by making an object more polished, compressed, or formal.
- If a conclusion appears stronger than its premises, the difference must have entered somewhere.
- The Rule: A weaker relation must not be silently promoted into a stronger one without an independent bridge principle.

NotebookLM

Scale 1: Formalism & The Epistemic Firewall



Formalism must preserve the status of its premises.

- ☑ If $A \vdash P$, this establishes the inferential link, not the premise A.
- ☑ Downstream mathematical rigor cannot upgrade the epistemic status of an initial assumption.
- ☑ The Discipline: A structurally honest presentation keeps definitions, derivations, and conjectures in visibly distinct strata.

NotebookLM

Scale 2: Enforcement in Code

CODE ARCHITECTURE

The Attestal Rust prototype implements **structural honesty** at the **type-system level.**

Result = Value + Generating History

A numerical result is irreversibly bound to its provenance record.

TERMINAL OUTPUT

```
> require_hardware()

[ERROR] claim rejected:
attempted 'real hardware
execution', but retained
provenance says 'simulation
(local prototype)'
```

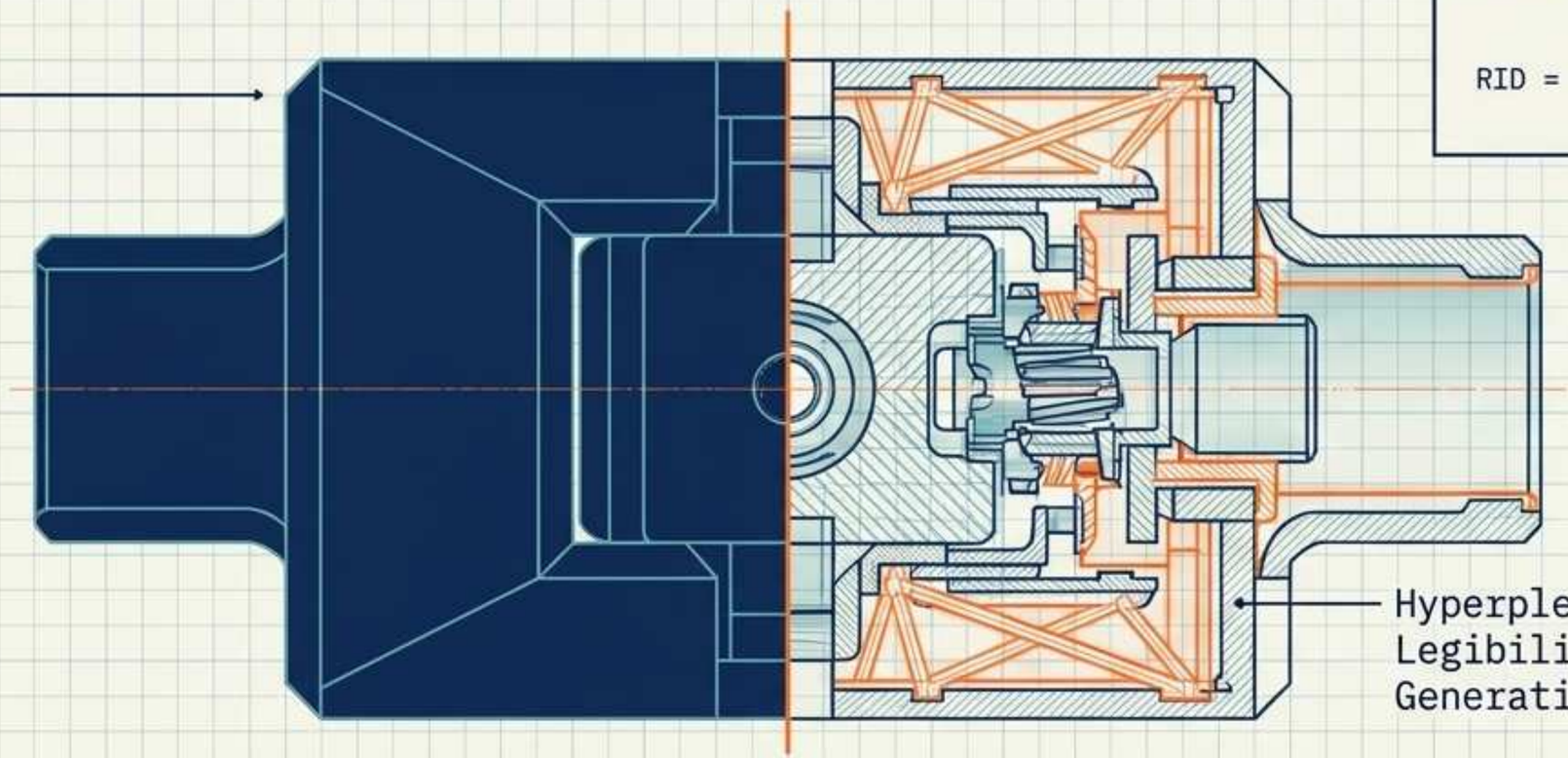
Takeaway: The type system does not evaluate if the claim is plausible; it evaluates if the retained history licenses it.

NotebookLM

Scale 3: Physical Artifacts & Generative Grammar

$$RID = \frac{I_{recoverable}}{I_{total}}$$

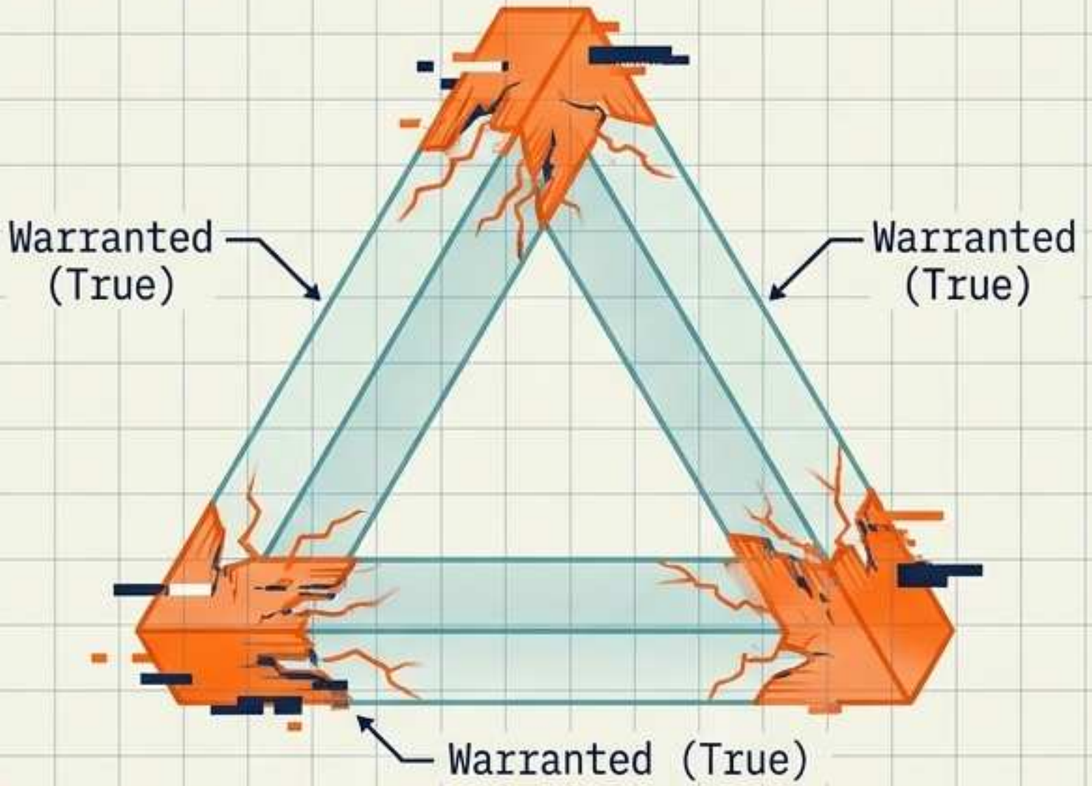
RID = Recoverable Information Density



- An artifact A possesses a generative grammar G.
- In opaque systems, the interpretive key K is externalized. The object functions, but loses the capacity to explain itself.
- A structurally honest artifact preserves access to its generative status, keeping the conditions of interpretation attached to the object itself.

NotebookLM

Scale 4: Compositional Dishonesty



$$\bigwedge_i W(c_i) \neq W(C(c_1, \ldots, c_n))$$

Every component can be independently warranted, yet their composition can be structurally dishonest.

- Ordering implies causation; juxtaposition implies comparison; shared scale implies commensurability.
- None of these relations are guaranteed by the truth of the components.

The TARTAN Principle: Local consistency does not guarantee global realizability. Passing internal tests is not adequacy to the world.

NotebookLM

Scale 5: Diagnostic Coordinates

Legibility Does Not Yet License Intervention.

Legibility ≠ Detection ≠ Diagnosis ≠ Repair Warrant



Structural text

- Each transition in the diagnostic sequence requires its own independent justification (a bridge principle).

Structural text

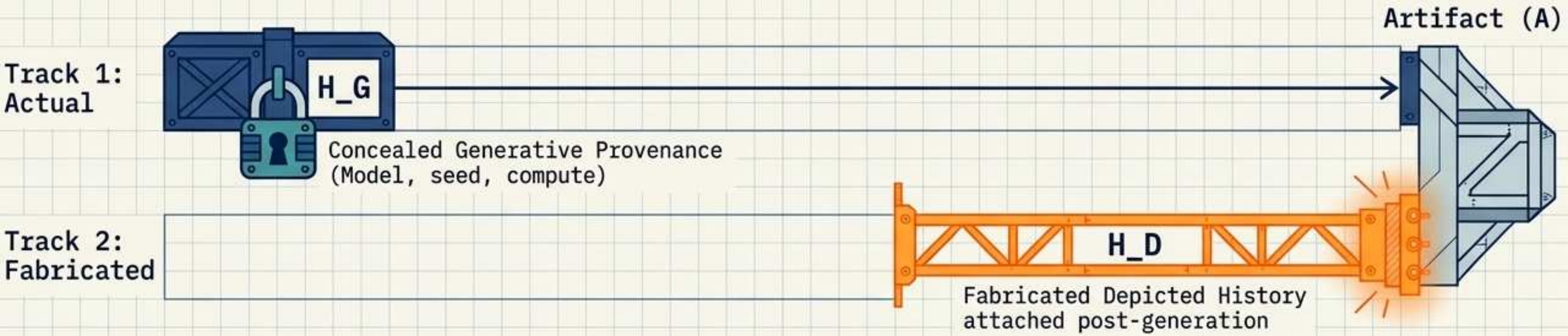
- A detected deviation is evidence of a discrepancy; it is not a license to correct it.

Structural text

- The capacity to see what could be changed is a separate achievement from possessing the structural warrant to change it.
- The null intervention (doing nothing) must remain a live, visible option.

NotebookLM

Scale 6: Synthetic Provenance & Time



Concealed Generative Provenance (H_G) + **Fabricated Depicted History (H_D)** = **Provenance Substitution**

Insight 1: Endpoint Limits

- **The Endpoint Cannot Testify:** $x(t) = y(t) \not\Rightarrow x \sim y$. A shared present state does not imply a shared history.

Insight 2: Computational vs. Depicted

- Synthetic artifacts possess computational provenance (**H_G**) but are attached to fabricated depicted histories (**H_D**) they never experienced.

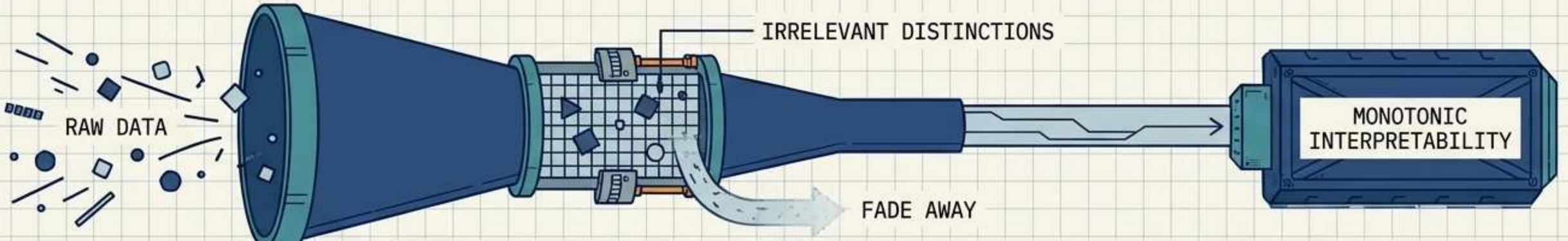
Insight 3: The Variant-Multiplicity Test

- When a ‘unique’ historical narrative has hundreds of mutually exclusive instantiations, it betrays a **generative template**, failing **global admissibility**.

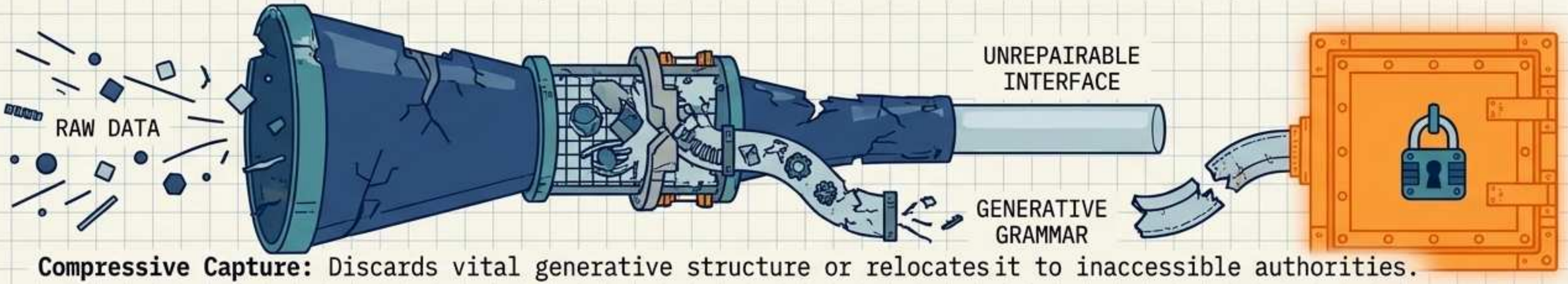
NotebookLM

The Right to Forget vs. Compressive Capture

An abstraction must earn the right to forget.



Honest Compression: Discards distinctions proven to be irrelevant. Allows monotonic refinement (experts can refine the coarse view without reversing it).

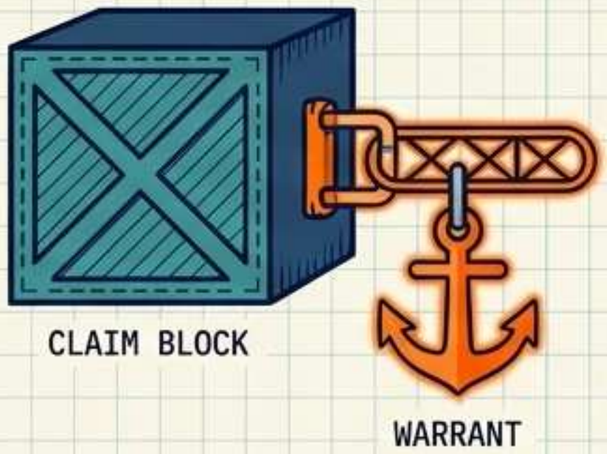


Compressive Capture: Discards vital generative structure or relocates it to inaccessible authorities.

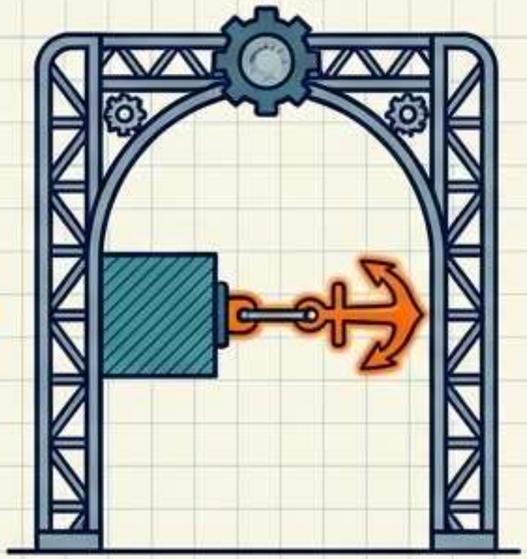
DIMENSIONAL POVERTY: Contraction of the control surface available to users for local repair. Interpretive power is redistributed, not just lost.

NotebookLM

The Auditability Funnel: Warrant Must Survive Transport

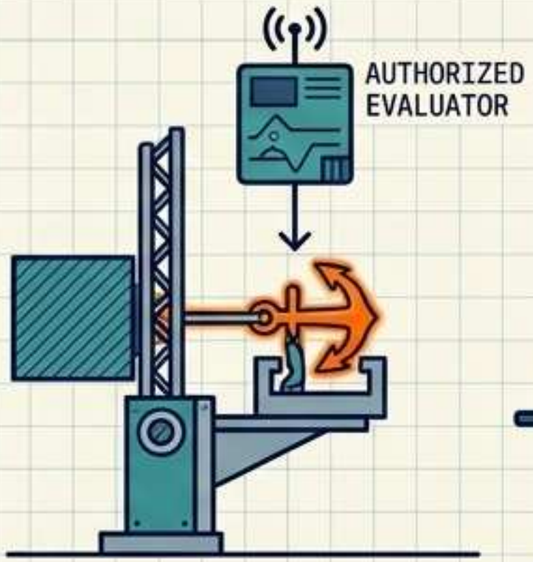


GATE 1: PRESERVATION



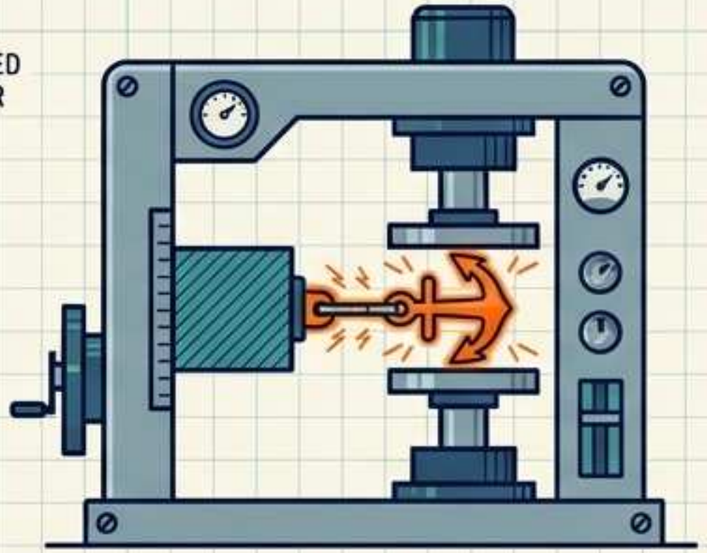
The anchor exists.

GATE 2: REACHABILITY



The anchor can be retrieved by an authorized evaluator.

GATE 3: AUDITABILITY



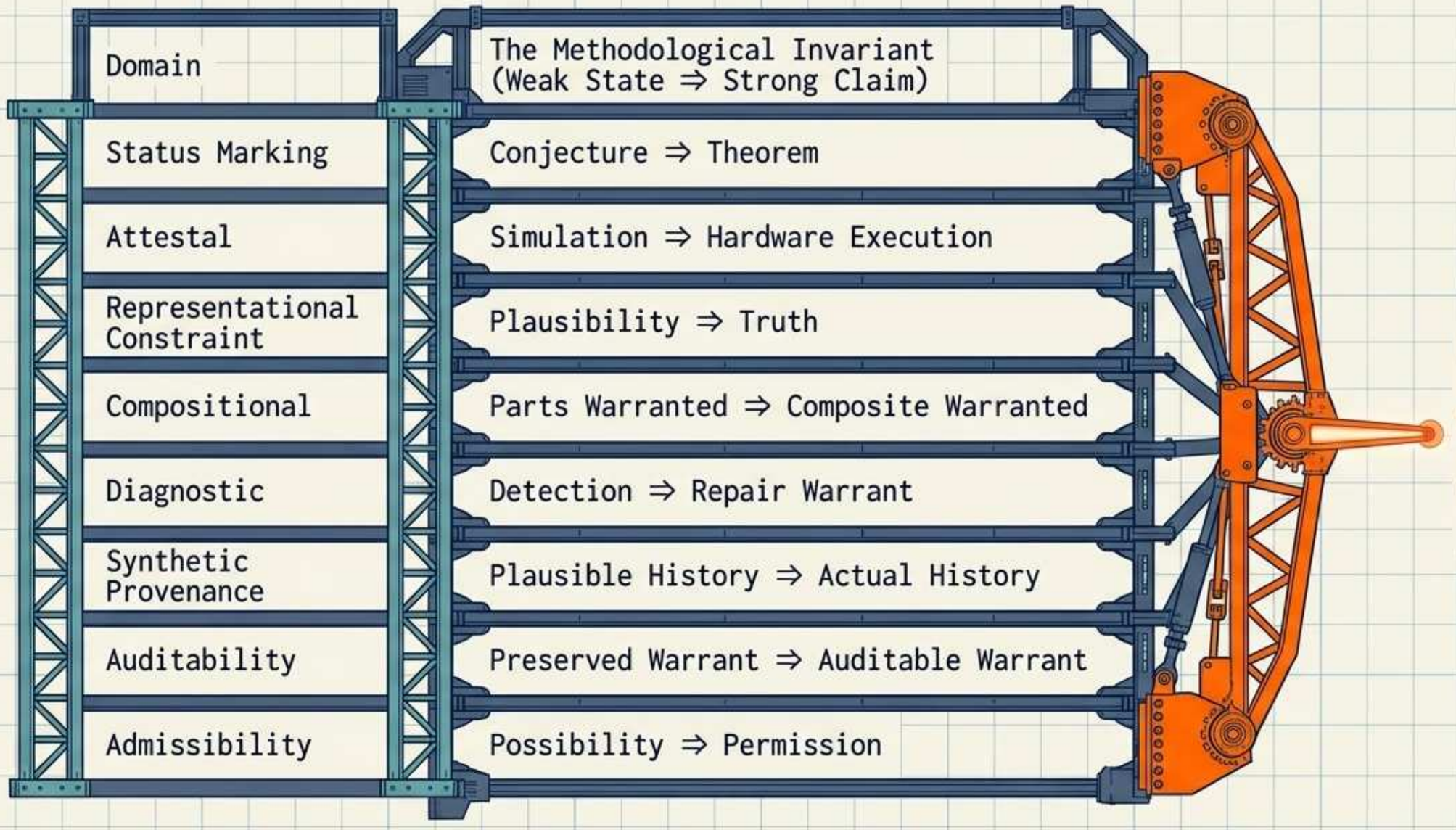
The connection is structured enough to test the promotion.

Preserved Warrant ⇏ Reachable Warrant ⇒ Auditable Warrant

A claim whose warrant predictably disappears on export, compression, quotation, or aggregation is only temporarily honest. Underdetermination must remain visible: “Unknown” is a positive epistemic state that prevents narrative substitution.

NotebookLM

The Methodological Invariant



No stronger relation without an explicit warrant for the promotion.

NotebookLM

The Architecture of Structural Honesty

The Autonomy of Refusal



STRUCTURAL HONESTY = NO DESTRUCTION OF RELEVANT WARRANT + NO FABRICATION OF ABSEN

NotebookLM